

WEEK 2 — LOOPS & SERIAL PORT CHEAT SHEET

PIR Motion Sensor | Data Science with ESP32 | Grade 8 | Keep this — refer every lesson!

1 WHILE LOOPS

```
# Loop forever — most common
while True:
    print("running")

# Loop while condition True
count = 0
while count < 10:
    count = count + 1

# Stop with break
while True:
    if score>100: break
```

STOP: Press Ctrl+C in terminal

2 FOR LOOPS

```
# Find ESP32 port
PORT = None
for port in list_ports.comports():
    if "CH340" in port.description:
        PORT = port.device

# Loop through list
readings = [45, 0, 72]
for score in readings:
    print(score)

# Fixed number of times
for i in range(10):
    print("Reading", i+1)
```

KEY: for = fixed times | while = forever

3 SERIAL PORT

```
# Open connection
ser = serial.Serial(PORT, 9600, time

# Read one line
line = ser.readline()

# Full one-liner
line = ser.readline().decode(
    "utf-8", errors="ignore").strip()
```

BAUD MUST MATCH: Arduino 9600 = Python 9600

4 STRING METHODS

<code>.decode("utf-8")</code>	Bytes to text
<code>errors="ignore"</code>	Skip bad bytes
<code>.strip()</code>	Remove whitespace/new
<code>.split(":")</code>	Cut text at character
<code>.split(":")[1]</code>	Take part after colon
<code>"text" in line</code>	Check if text exists
<code>line.lower()</code>	Convert to lowercase

5 TRY / EXCEPT

```
# Basic — skip any error
try:
    score = int(line)
except:
    pass # move on silently

# Specific error type
try:
    val = float(line.split(":")[1])
except ValueError:
    pass # only catch ValueError

# Check first, try second
if line: # not empty?
    try:
        score = int(line)
    except: pass
```

USE WHEN: Converting text to number from sensor

6 AUTO PORT DETECTION

```
import serial.tools.list_ports

PORT = None
for port in list_ports.comports():
    if "CH340" in port.description:
        PORT = port.device
        print("Found:", PORT)

if PORT is None:
    print("ESP32 not found!")
else:
    ser = serial.Serial(PORT, 9600)
```

FULL PIR SCRIPT

```
import serial
import serial.tools.list_ports
from datetime import datetime

BAUD = 9600
PORT = None

for port in list_ports.comports():
    if "CH340" in port.description:
        PORT = port.device

print("Found:", PORT)
ser = serial.Serial(PORT, BAUD, timeo

print("Reading PIR... Ctrl+C to stop
print()

while True:
    line = ser.readline().decode(
        "utf-8", errors="ignore").str
    if line:
        try:
            score = int(line)
            now = datetime.now()
            now = now.strftime("%H:%M)
            if score > 0:
                status = "Motion Dete
            else:
                status = "No Motion"
            print(f" {now} {score:<
        except:
            pass
```

COMMON ERRORS

UnicodeDecodeError

Fix: Add errors="ignore"

TypeError: NoneType

Fix: Check if line: before int()

AttributeError

Fix: Check if PORT: before Serial()

ValueError: literal

Fix: Wrap int() in try/except

THE 3 GOLDEN RULES — EVERY LESSON:

1. Upload matching Arduino code to ESP 32 FIRST
2. Close Arduino IDE completely before running Python
3. Only ONE program can use COM5 at a time — never both!